

Acute effects of high-fat meals enriched with walnuts or olive oil on postprandial endothelial function.



OBJECTIVES: We sought to investigate whether the addition of walnuts or olive oil to a fatty meal have differential effects on postprandial vasoactivity, lipoproteins, markers of oxidation and endothelial activation, and plasma asymmetric dimethylarginine (ADMA). **BACKGROUND:** Compared with a Mediterranean diet, a walnut diet has been shown to improve endothelial function in hypercholesterolemic patients. We hypothesized that walnuts would reverse postprandial endothelial dysfunction associated with consumption of a fatty meal. **METHODS:** We randomized in a crossover design 12 healthy subjects and 12 patients with hypercholesterolemia to 2 high-fat meal sequences to which 25 g olive oil or 40 g walnuts had been added. Both test meals contained 80 g fat and 35% saturated fatty acids, and consumption of each meal was separated by 1 week. Venipunctures and ultrasound measurements of brachial artery endothelial function were performed after fasting and 4 h after test meals. **RESULTS:** In both study groups, flow-mediated dilation (FMD) was worse after the olive oil meal than after the walnut meal ($p = 0.006$, time-period interaction). Fasting, but not postprandial, triglyceride concentrations correlated inversely with FMD ($r = -0.324$; $p = 0.024$). Flow-independent dilation and plasma ADMA concentrations were unchanged, and the concentration of oxidized low-density lipoproteins decreased ($p = 0.051$) after either meal. The plasma concentrations of soluble inflammatory cytokines and adhesion molecules decreased ($p < 0.01$) independently of meal type, except for E-selectin, which decreased more ($p = 0.033$) after the walnut meal. **CONCLUSIONS:** Adding walnuts to a high-fat meal acutely improves FMD independently of changes in oxidation, inflammation, or ADMA. Both walnuts and olive oil preserve the protective phenotype of endothelial cells.